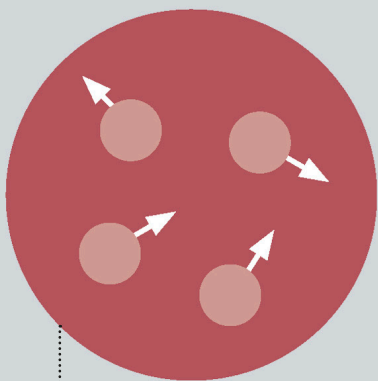
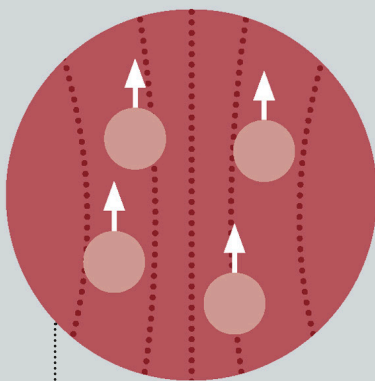




**RANDOM
POSITION**

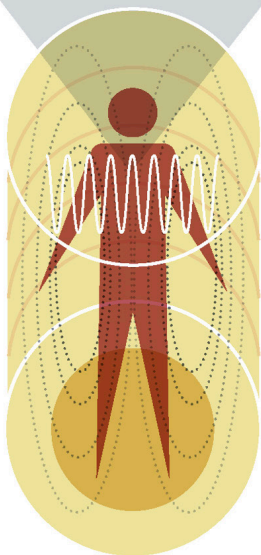
With no external magnetic field, protons spin inside the body with the axes of their own magnetic fields randomly aligned.



**MOSTLY
ALIGNED**

When a powerful external magnetic field is applied, the axis of each proton's magnetic field lines up (parallel) with this magnetic field.

MRI SCANNER



Quantum scanner

When the spin of a single proton is measured, it can be in one of two states: parallel or antiparallel. During an MRI scan, protons switch from parallel to antiparallel and back again.

"It was eerie. I saw myself in that machine. I never thought my work would come to this."

Isidor Isaac Rabi,
whose work led to the MRI